# Aluffi, Chapter-0

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**Example. aluffi** Define $\mathsf{C}$ as follows: Let $S$ be any set, and define a category $\mathsf{C} := \hat{\mathsf{S}}$ by setting

- $\mathsf{Obj}(\hat{\mathsf{S}}) = \mathscr{P}(S)$
- for any pair of objects $A, B \in \mathsf{Obj}(\hat{\mathsf{S}})$ let $\mathsf{Hom}_{\mathsf{C}}(A, B)$ be the pair $(A, B)$ if $A \subseteq B$, and let $\mathsf{Hom}_{\mathsf{C}}(A, B) = \varnothing$ otherwise.

Indeed, notice that since every $A \in \mathsf{Obj}(\hat{\mathsf{S}})$ satisfies $A \subseteq A$, every $\mathsf{Hom}_{\mathsf{C}}(A, A)$ is nonempty with the identity morphism $1_A$ being the pair $(A, A)$.

Next, if there is morphism $A \longrightarrow B$ and a morphism $B \to C$ then this means

that there is a morhpism $A \to C$ which is just the pair $(A, C)$. Indeed, since $A \subseteq B$ and $B \subseteq C$ implies that $A \subseteq C$, so $\mathsf{Hom}_{\mathsf{C}}(A, C)$ is nonempty.

Next, if there are morphisms $f : A \longrightarrow B$, $g : B \longrightarrow C$ and $h : C \longrightarrow D$ for objects $A, B, C, D \in \mathsf{Obj}(\hat{\mathsf{S}})$ then the morphism $hg = (B, D)$, $f = (A, B)$, $gf = (A, C)$ and so $(hg)f = (A, D) = h(gf)$. So, associativity also holds.

That $A \xrightarrow{f} B$ satisfies $f1_A = f$ and $1_B f = f$ is obvious because $1_A = (A, A)$ and $f = (A, B)$ so the morphism $f1_A$ is just $(A, B)$ while the morphism $1_B f$ is also just $(A, B)$.

**Example. aluffi** Fix a set $S$ and a relation $\sim$ that is reflexive and transitive. Then define a category $\mathsf{C}$ as such.

- the objects of $\mathsf{C}$ are elements of $S$.
- if $a, b \in \mathsf{Obj}(\mathsf{C})$ then $\mathsf{Hom}_{\mathsf{C}}(a, b)$ is a set containing the element $(a, b)$ if $a \sim b$ and $\mathsf{Hom}_{\mathsf{C}}(a, b) = \varnothing$ otherwise.

By reflexivity, $\mathsf{Hom}(a, a)$ is always nonempty because it contains the element $1_a = (a, a)$ always.

If $f : a \to b$ and $g : b \to c$ are any two morphisms then this means that $a \sim b$ and $b \sim c$. By transitivity, $a \sim c$ and so $gf : A \to C$ is a well defined morphism. In fact it is the element $(a, c)$. That is, $\mathsf{Hom}(a, c) = \{gf\} = \{(a, c)\}$.

If $f : a \to b$, $g : b \to c$ and $h : c \to d$ are morphisms in $\mathsf{C}$ then
$hg = (b, d)$ and $gf = (a, c)$ so $(hg)f = (a, d) = h(gf)$.

Next, for any morphism $f : a \to b$, $1_b f = (a, b) = f$ and $f1_a = (a, b) = f$.

An interesting fact about this category is that given any morphism $f : a \to b$, take any object $c$ and a pair of morphisms $\alpha', \alpha'' : c \to a$. If $c, \alpha', \alpha''$ are so that

$$f\alpha' = f\alpha''$$

then this implies that $\alpha' = \alpha''$. In fact $\alpha' = \alpha''$ is always true because $\mathsf{Hom}_{\mathsf{C}}(c, a)$ is always a singleton or empty. This discussion elucidates the fact that every morphism in this category is a *monomorphism*.

In fact, take any morphism $f : a \to b$ and fix any object $c \in \mathsf{Obj}(\mathsf{C})$. Take any pair of morphisms $\beta', \beta'' : b \to c$ so that $\beta' f = \beta'' f$. Then, $\beta' = \beta''$.

Once again, $\beta' = \beta''$ always holds for the same reason as above. Also, $\beta' f = \beta'' f$ also always holds because $\beta' f = (b, c)$ and so is $\beta'' f$. Every morphism in this category is an *epimorphism.*

**Example. aluffi** Let $\mathsf{C}$ be a category and fix $A \in \mathsf{Obj}(\mathsf{C})$. We will define another category from $\mathsf{C}$, denoted $\mathsf{C}_A$ as

- objects of $\mathsf{C}_A$ are morphisms from objects of $\mathsf{C}$ to $A$. That is, elements of $\mathsf{Obj}(\mathsf{C}_A)$ are of the form $Z \xrightarrow{f} A$ for any object $Z \in \mathsf{Obj}(\mathsf{C})$.
- morphisms between two objects $f, g \in \mathsf{Obj}(\mathsf{C}_A)$ are diagrams comprising of objects of $\mathsf{C}$ and the appropriate morphisms between them.

Given two objects $f\colon Z_1 \to A$ and $g\colon Z_2 \to A$ in $\mathsf{C}_A$ a morphism between $f, g$ in $\mathsf{C}_A$ is a *commutative diagram* of the form

$$\begin{tikzcd}
Z_1 \arrow[rr, dashed, "\sigma"] \arrow[dr, "f"'] & & Z_2 \arrow[dl, "g"] \\
& A &
\end{tikzcd}$$

We will show that this construction of $\mathsf{C}_A$ forms a category.

First, we will show that $\mathsf{End}_{\mathsf{C}_A}(f)$ for any $f \in \mathsf{Obj}(\mathsf{C}_A)$ is nonempty and contains the identity morphism by noticing that the diagram

$$\begin{tikzcd} Z \arrow[rr, dashed, "1_Z"] \arrow[dr, "f"'] & & Z \arrow[dl, "f"] \\ & A & \end{tikzcd}$$

where $1_Z$ is the identity morphism for the object $Z \in \mathsf{Obj}(\mathsf{C})$. This diagram is the element $1_f \in \mathsf{Hom}_{\mathsf{C}_A}(f, f)$.

Take any two morphisms $f_1 \to f_2$ and $f_2 \to f_3$ in $\mathsf{C}_A$ next.

We will show that this defines a new morphism which is the composition of these two morphisms. For this we concatenate the commutative diagrams $f_1 \to f_2$ and $f_2 \to f_3$ side by side.

$$\begin{tikzcd} Z_1 \arrow[r, "\sigma"] \arrow[dr, "f_1"'] & Z_2 \arrow[r, "\tau"] \arrow[d, "f_2"] & Z_3 \arrow[dl, "f_3"] \\ & A & \end{tikzcd}$$

We will show that the following diagram contained in this diagram also commutes.

$$\begin{tikzcd} Z_1 \arrow[rr, "\tau\sigma"] \arrow[dr, "f_1"'] & & Z_3 \arrow[dl, "f_3"] \\ & A & \end{tikzcd}$$

Indeed, since $f_1 = f_2\sigma$ and $f_2 = f_3\tau$ from the previous diagram due to commutativity so $f_1 = f_3\tau\sigma = f_3(\tau\sigma)$ which proves that this diagram also commutes, therefore, the composition of the morphisms $f_1 \to f_2$ and

$f_2 \to f_3$ is also a commutative diagram which describes the morphism $f_1 \to f_3$ between objects $f_1$ and $f_3$ of the category $\mathsf{C}_A$.

Next, suppose there are three morphisms $f_1 \to f_2 \to f_3 \to f_4$ of objects $f_1, f_2, f_3$ and $f_4$ in $\mathsf{Obj}(\mathsf{C}_A)$. That means there are three diagrams as below

$$\begin{tikzcd}
Z_1 \arrow[r, "\alpha"] \arrow[dr, "f_1"'] & Z_2 \arrow[r, "\beta"] \arrow[d, "f_2"'] & Z_3 \arrow[dl, "f_3"] \arrow[d, "\gamma"] \\
 & A & Z_4 \arrow[l, "f_4"]
\end{tikzcd}$$

Since $f_1 \to f_2$ is a morphism of objects $f_1$ and $f_2$ in $\mathsf{C}_A$, the associated diagram above for the two morphisms is commutative. The commutativity relation is $f_1 = f_2\alpha$. Similarly, we argue for the other three diagrams and find the following equalities by commutativity:

$$f_2 = f_3\beta \text{ and}$$
$$f_3 = f_4\gamma$$

Let us set $\sigma := f_1 \to f_2, \tau := f_2 \to f_3, \delta := f_3 \to f_4$ as our diagrams. Then, the diagram $\delta\tau$ is as

$$\begin{tikzcd}
Z_2 \arrow[d, "f_2"'] \arrow[dr, "\gamma\beta"] & \\
A & Z_4 \arrow[l, "f_4"]
\end{tikzcd}$$

and the diagram $(\delta\tau)\sigma$ is

$$\begin{tikzcd}
Z_1 \arrow[d, "f_1"'] \arrow[dr, "\gamma\beta\alpha"] & \\
A & Z_4 \arrow[l, "f_4"]
\end{tikzcd}$$

Next, the diagram $\tau\sigma$ is

$$\begin{tikzcd}
Z_1 \arrow[r, "\beta\alpha"] \arrow[d, "f_1"'] & Z_3 \arrow[dl, "f_3"] \\
A &
\end{tikzcd}$$

and finally the diagram for $\delta(\tau\sigma)$ is

$$\begin{tikzcd}
Z_1 \arrow[d, "f_1"'] \arrow[dr, "\gamma\beta\alpha"] & \\
A & Z_4 \arrow[l, "f_4"]
\end{tikzcd}$$

So, $\delta(\tau\sigma) = (\delta\tau)\sigma$ proves associativity.

That given any morphism $\alpha : f_1 \to f_2$ of $\mathsf{C}_A$, $1_{f_2}\alpha = \alpha$ and $\alpha 1_{f_1} = \alpha$ is obvious from the same kinds of steps taken above.

**Example. aluffi**

For the category $\mathsf{C}$ associated to the set of integers with the $\leqslant$ being $\sim$ as in Eg-3.3.

fix an object $A = 3$. Then, any object of $\mathsf{C}_A$ is a pair $(x, 3) \in \mathbb{Z} \times \mathbb{Z}$ with $x \leqslant 3$. There is a morphism

$(m, 3) \to (n, 3)$ if and only if $m \leqslant n$.

We learned about the slice category, now let us see an example of a category that is dual to this - the coslice category.

**Example 1. aluffi** For a concrete example, take the category $\mathsf{Set}$ for which $\mathsf{Obj}(\mathsf{Set})$ is the class of all sets. Fix a singleton $\{*\}$ in $\mathsf{Set}$. Next, we define a new category $\mathsf{Set}^*$ whose objects are all the morphisms $f : \{*\} \to A$ for any object $A$ of $\mathsf{Set}$. Any object in $\mathsf{Set}$ may be identified as a pair $(S, s)$ where $s = f(*)$ because the information of any object in $\mathsf{Set}^*$ comprises of the choice of the nonempty set $S$ and of an element $s \in S$ that $f$ maps $*$ to in $S$.

A morphism between two objects $(S, s)$ and $(T, t)$ in $\mathsf{Set}^*$ corresponds to a map $\sigma : S \to T$ *such that* $\sigma(s) = t$.

Objects in $\mathsf{Set}^*$ are called *pointed sets*.

**Exercise. aluffi** Let $\mathsf{C}$ be a category. Consider a structure $\mathsf{C}^{\mathsf{op}}$ with

- $\mathsf{Obj}(\mathsf{C}^{\mathsf{op}}) \coloneqq \mathsf{Obj}(\mathsf{C})$;
- for a pair of objects $A, B \in \mathsf{Obj}(\mathsf{C}^{\mathsf{op}})$, $\mathsf{Hom}_{\mathsf{C}^{\mathsf{op}}}(A, B) \coloneqq \mathsf{Hom}_{\mathsf{C}}(B, A)$

that is, given any pair of objects $A, B$ of the category $\mathsf{C}^{\mathsf{op}}$, arrows from $A$ to $B$ are arrows from $B$ to $A$ in $\mathsf{C}$.

**Solution.** Take any pair of objects $A, B \in \mathsf{Obj}(\mathsf{C}^{\mathsf{op}})$. Define

$$\circledcirc_{\mathsf{C}^{\mathsf{op}}} : \mathsf{Hom}_{\mathsf{C}^{\mathsf{op}}}(A, B) \times \mathsf{Hom}_{\mathsf{C}^{\mathsf{op}}}(B, C) \to \mathsf{Hom}_{\mathsf{C}^{\mathsf{op}}}(A, C)$$

by the following rule:

$$g \circledcirc_{\mathsf{C}^{\mathsf{op}}} f \coloneqq f \circledcirc g$$

where the $\circledcirc$ in the right hand side is the composition of $f$ and $g$ in $\mathsf{C}$.

Since $\mathsf{Hom}_{\mathsf{C}}(A, A)$ is nonempty for every $A \in \mathsf{Obj}(\mathsf{C})$ and objects of $\mathsf{C}^{\mathsf{op}}$ are simply objects in $\mathsf{C}$, $\mathsf{Hom}_{\mathsf{C}^{\mathsf{op}}}(A, A)$ is also nonempty and contains the same $1_A \in \mathsf{Hom}_{\mathsf{C}}(A, A)$.

The morphism $f \circledcirc_{\mathsf{C}^{op}} g$ is an element of $\mathsf{Hom}_{\mathsf{C}^{op}}(A, C)$ because $f \circ g$ is an element of $\mathsf{Hom}_{\mathsf{C}}(C, A)$. The well definedness of the map $\circledcirc_{\mathsf{C}^{op}}$follows immediately from the well definedness of $\circledcirc$. Just take elements $f, f' \in \mathsf{Hom}_{\mathsf{C}}(B, A)$ and $g, g' \in \mathsf{Hom}_{\mathsf{C}}(C, B)$ with $f = f'$, $g = g'$ and then notice that

$$\begin{aligned} g \circledcirc_{\mathsf{C}^{op}} f =& f \circledcirc g \\ =& f' \circledcirc g' \\ =& g' \circledcirc_{\mathsf{C}^{op}} f' \end{aligned}$$

Next, take the morphisms $f : A \to B, g : B \to C, h : C \to D$ in $\mathsf{C}^{op}$.
Then,

$$\begin{aligned} (h \circledcirc_{\mathsf{C}^{op}} g) \circledcirc_{\mathsf{C}^{op}} f &= f \circledcirc (g \circledcirc h) \\ &= (f \circledcirc g) \circledcirc h \\ &= h \circledcirc_{\mathsf{C}^{op}} \left(g \circledcirc_{\mathsf{C}^{op}} f\right) \end{aligned}$$

showing that the composition is associative.
Finally, pick a morphism $f \in \mathsf{Hom}_{\mathsf{C}^{op}}(A, B)$. Then,

$$\begin{aligned} 1_B \circledcirc_{\mathsf{C}^{op}} f =& f \circledcirc 1_B \\ =& fp \end{aligned}$$

and

$$\begin{aligned} f \circledcirc_{\mathsf{C}^{op}} 1_A =& 1_A \circledcirc f \\ =& f \end{aligned}$$

This completes our construction and proof of construction of the opposite category.

*Remark.* We choose to indicate the morphism $f \in \mathsf{Hom}_{\mathsf{C}^{op}}(A, B)$ by just the same symbol when we consider it in $\mathsf{Hom}_{\mathsf{C}}(B, A)$. This might not be a bit tricky to deal with initially but it does make writing the proof quite more convenient. ■

**Exercise. aluffi** If $A$ is a finite set, how large is $\mathsf{End}_{\mathsf{Set}}(A)$?

**Solution.** The morphisms in $\mathsf{Set}$ are simply functions between the objects (sets) of $\mathsf{Set}$. So we are simply counting the number of functions from a finite set to itself.

I claim that for a finite set $A$ of $n$ elements, there are $n^n$ functions $A \to A$. Since bijections compose to give bijections, it suffices to prove the statement for a set of the form $\{1, \ldots, n\}$.

We notice that there is a bijection $\{1, \ldots, n\}^{\{1,\ldots,n\}} \to \prod_{k=1}^{n}\{1, \ldots, n\}$. Indeed, the map

$$f \mapsto (f(i))_{i=1}^{n}$$

is the obvious candidate. By the product rule of combinatorics, we see that $\prod_{k=1}^{n}\{1, \ldots, n\}$ has cardinality $n^n$ and we are done. ■

**Exercise. aluffi** Formulate precisely what it means to say that $1_a$ is an identity with respect to composition in Eg-3.3.

**Solution.** Take any pair of objects $a, b \in \mathsf{Obj}(\mathsf{C})$.If $\mathsf{Hom}_{\mathsf{C}}(a, b)$ is empty, then the statement is trivially true. If not then pick $(a, b) \in \mathsf{Hom}_{\mathsf{C}}(a, b)$. It is clear that we have

$$(a, b)1_a = (a, b)(a, a) = (a, b)$$

which means that $1_a$ is the identity with respect to composition. We can do similar for $1_b$. ■

**Exercise. aluffi** Can we define a category in the style of Eg-3.3 using the relation $<$ on $\mathbb{Z}$.

**Solution.** Every $\mathsf{Hom}_{\mathsf{C}}(a, a)$ needs to be nonemtpy. If we try to define a category using < we miss out on $a \sim a$ which does not give us any obvious candidate of $1_a \in \mathsf{Hom}_{\mathsf{C}}(a, a)$. Shortly put, $<$ is not reflexive. ■

**Exercise. aluffi** Define in what sense the category constructed in Eg-3.4 coincides with that in Eg-3.3.

**Solution.** Given a set $S$, define the relation $\sim$ on $\mathscr{P}(S)$ as such: any two subsets $A, B$ of $S$ are related by $\sim$, as in $A \sim B$ if and only if $A \subseteq B$. The rest is obvious.

Simply put, the $\subseteq$ relation is a preorder. That is, $A \subseteq A$ for every susbet $A$ of $S$ and $A \subseteq B, B \subseteq C$ means $A \subseteq C$ for any subsets $A, B, C$ of $S$. ■

**Exercise. aluffi** A *subcategory* $\mathsf{C}'$ of a category $\mathsf{C}$ consists of a collection of objects of $\mathsf{C}$, with morphisms $\mathsf{Hom}_{\mathsf{C}'}(A, B) \subseteq \mathsf{Hom}_{\mathsf{C}}(A, B)$ for all objects $A, B \in \mathsf{Obj}(\mathsf{C}')$, such that identities and compositions in $\mathsf{C}$ make $\mathsf{C}'$ into a category. A subcategory $\mathsf{C}'$ is full if $\mathsf{Hom}_{\mathsf{C}'}(A, B) = \mathsf{Hom}_{\mathsf{C}}(A, B)$ for all $A, B \in \mathsf{Obj}(\mathsf{C}')$. Construct a category of *infinite sets* and explain how it may be viewed as a full subcategory of $\mathsf{Set}$.

**Solution.** We define a category $\mathsf{InfSet}$ as such:

- the objects of $\mathsf{InfSet}$ are simply the infinite sets ni $\mathsf{Obj}(\mathsf{Set})$, that is $\mathsf{Obj}(\mathsf{InfSet}) \subseteq \mathsf{Obj}(\mathsf{Set})$

- given any two infinite sets $A, B \in \mathsf{Obj}(\mathsf{InfSet})$ we define the set of morphisms $A \to B$ by $\mathsf{Hom}_{\mathsf{InfSet}}(A, B) := \mathsf{Hom}_{\mathsf{Set}}(A, B)$.

That these constructions make $\mathsf{InfSet}$ into a subcategory of $\mathsf{Set}$ is obvious. It has the same identity and compositions as $\mathsf{Set}$.

That it is a full subcategory of $\mathsf{Set}$ is by definition. ■

**Exercise. aluffi** Define a category $\mathsf{V}$ by taking $\mathsf{Obj}(\mathsf{V}) := \mathbb{N}$ and letting $\mathsf{Hom}_{\mathsf{V}}(n, m) :=$ the set of $m \times n$ matrices with real entries, for all $n, m \in \mathbb{N}$. Use product of matrices to define composition. Does this category *feel* familiar?

**Solution.** Given objects $j, k, l \in \mathsf{Obj}(V)$, define morphisms $M_{k\times j} \in \mathsf{Hom}_V(j,k)$, $N_{l\times k} \in \mathsf{Hom}_V(k,l)$ where $M_{k\times j}$ and $N_{l\times k}$ are $k \times j$ and $l \times k$ shaped matrices of reals respectively. Define composition

$\mathsf{Hom}(j,k) \times \mathsf{Hom}(k,l) \to \mathsf{Hom}(j,l)$

by the rule

$M_{k\times j} \circ N_{l\times k} \coloneqq N_{l\times k} M_{k\times j}$

where the product on the right hand side is matrix multiplication which is well defined for $l \times k$ and $k \times j$ matrices. The result is a matrix $l \times j$ which is an element of $\mathsf{Hom}_V(j,l)$.

Given any morphism $M_{n\times m}$, consider the morphism $1_n \in \mathsf{Hom}(n,n)$. Notice that $1_n M_{n\times m} = M$ and $M_{n\times m} 1_m = M$ which shows that $V$ is a category.

This is very familiar to the category $\mathsf{Vect}$ of all finite dimensional $k-$vector spaces. The objects are finite dimensional spaces and the morphisms are linear maps. The spaces can be identified by their dimension which is a natural number and the linear maps can be identified by matrices. This suggests there is a sense of *isomorphism* or *sameness* between these two categories. Such ideas have not been covered yet in the book so we will not dig further into them for now.

The matrix of 0 rows and 0 columns can represent a linear operator on the trivial vector space $\{0\}$ which has dimension 0. ■

**Exercise. aluffi** Define carefully objects and morphisms in Eg-3.7, and draw the diagram corresponding to composition.

**Solution.** Suppose $\mathsf{C}$ is given as a category. Then, fix some object $S \in \mathsf{Obj}(\mathsf{C})$. Define the new category $\mathsf{C}^S$ as follows:

- objects in $\mathsf{C}^S$ are morphisms $S \to A$ for objects $A \in \mathsf{Obj}(\mathsf{C})$,
- morphisms between objects $f_{1,} f_2$ in $\mathsf{C}^s$ are diagrams as follows:

$$\begin{tikzcd} & S \arrow[dl, "f_1"'] \arrow[dr, "f_2"] & \\ Z_1 & & Z_2 \arrow[ll, dashed, "\sigma"] \end{tikzcd}$$

with $f_1 = \sigma f_2$; that is to say that the diagrams are commutative.

Here, $Z_1, Z_2 \in \mathsf{Obj}(\mathsf{C})$.

Given any object $f : S \to Z \in \mathsf{Hom}_{\mathsf{C}}(S, Z)$ of $\mathsf{C}^S$ where $Z \in \mathsf{Obj}(\mathsf{C})$ there is a diagram

$$\begin{tikzcd} & S \arrow[dl, "f"'] \arrow[dr, "f"] & \\ Z \arrow[rr, dashed, "1_f"] & & Z \end{tikzcd}$$

which obviously commutes and therefore, if we denote this diagram by $1_f$ then $1_f \in \mathsf{Hom}_{\mathsf{C}^S}(f, f)$ for every $f \in \mathsf{Obj}(\mathsf{C}^S)$.

Given two morphisms $f_1 \to f_2 \to f_3$ in $\mathsf{C}^S$, we will define their composition by the following commutative diagram:

$$\begin{tikzcd} Z_1 & Z_2 \arrow[l, "\alpha"'] & Z_3 \arrow[l, "\beta"'] \\ & A \arrow[ul, "f_1"] \arrow[u, "f_2"] \arrow[ur, "f_3"'] & \end{tikzcd}$$

This diagram indeed commutes because $f_1 = \alpha f_2, f_2 = \beta f_3$ so $f_1 = \alpha\beta f_3$

which gives the commutative diagram

$$\begin{tikzcd}
Z_1 & & Z_3 \arrow[ll, "\alpha\beta"'] \\
& A \arrow[ul, "f_1"] \arrow[ur, "f_3"'] &
\end{tikzcd}$$

which describes a morphism $f_1 \to f_3$ proving that compositions exist and are well defined.

That $1_f$ is the identity with respect to composition can be proved from this easily. We shall only prove the associativity of compositions next.

For this, take three morphisms $f_1 \to f_2 \to f_3 \to f_4$ of objects $f_1, f_2, f_3, f_4$ in $\mathsf{C}^S$.

Consider the diagram

$$\begin{tikzcd}
Z_1 & Z_2 \arrow[l, "\alpha"'] & Z_3 \arrow[l, "\beta"'] \\
& A \arrow[ul, "f_1"] \arrow[u, "f_2"] \arrow[ur, "f_3"] \arrow[r, "f_4"] & Z_4 \arrow[u, "\gamma"]
\end{tikzcd}$$

Next, we will label the morphisms $\sigma : f_3 \to f_4, \tau : f_2 \to f_3, \lambda : f_1 \to f_2$ and look first at $(\sigma\tau)\lambda$. It is clear that the composition $\sigma\tau$ has the diagram without $Z_3$ and $f_3$ in the middle with an arrow $Z_4 \to Z_2$ that is the morphism $\beta\gamma$. This, composed with $f_1$ gives us a diagram with no $Z_2$ and $f_2$ in the middle with a map $Z_4 \to Z_1$ given with the arrow $\alpha\beta\gamma$.

Next, we look at $\sigma(\tau\lambda)$. Clearly, $\tau\lambda$ gives us the diagram without $Z_2$ and $f_2$ in the middle with a morphism from $Z_3 \to Z_1$ given as $\alpha\beta$. Composing this new diagram with the diagram $\sigma$ erases the $f_3$ and $Z_3$ in the middle giving an arrow $Z_4 \to Z_1$ which is the morphism $\alpha\beta\gamma$. This completes the proof. ■

**Exercise. aluffi** Define a morphism between multisets, obtaining a category $\mathsf{MSet}$ containing a copy of $\mathsf{Set}$ as a full subcategory. Which objects in $\mathsf{MSet}$ determine ordinary multisets defined in 2.2 and how? What does a morphsim of multiset look like here?

**Solution.** TODO

**Exercise. aluffi** Since the objects of a category $\mathsf{C}$ are not necessarily sets, it is not clear how to make sense of a notion of 'subobjects' in general. In some situations, it does make sense to talk about subobjects, and the subobjects of any given object $A$ in $\mathsf{C}$ are in one-to-one correspondence with the morphism $A \to \Omega$ for a fixed, special object $\Omega$ of $\mathsf{C}$, called a subobject classifier. Show that $\mathsf{Set}$ has a subobject classifier. ■

**Solution.** Fix $\Omega \coloneqq \{0, 1\} \in \mathsf{Obj}(\mathsf{Set})$. Clearly every subobject of a set $A$ in $\mathsf{Obj}(\mathsf{Set})$ should be its subset and every subset $B$ of $A$ can be identified by its indicator $B \to \Omega, b \mapsto 1$ if element $b$ of $A$ satisifes $b \in B$ and $b \mapsto 0$ otherwise. The set of subobjects of $A$ should be the $\mathscr{P}(A)$ so we will establish a bijection $f : \mathscr{P}(A) \to \{0, 1\}^A$.

This bijection will be given by the map $B \mapsto \mathbb{1}_B$ where $\mathbb{1}_B$ is the indicator of the set $B$ in $A$. This map is injective because $\mathbb{1}_C = \mathbb{1}_B$ says an element $x \in A$ satisfies $x \in C$ if and only if $x \in B$. So, $C = B$.

For surjectivity, suppose we are given a map $g \in \{0, 1\}^A$. We will find a subset $B \subseteq A$ so that $f(B) = g$. Set $B \coloneqq g^{-1}(1)$. Then, $f(B) = \mathbb{1}_B$. If we show that $\mathbb{1}_B = g$ on $A$ then we are done. So, take any $a \in A$. If $a \in B$ then $\mathbb{1}_B(a) = 1$ and $g(a) = 1$ because $a \in g^{-1}(1)$. If $a \in A \setminus B$ then $\mathbb{1}_B(a) = 0$ while $g(a) = 0$ because $a \notin g^{-1}(1)$ and every point of $A$ must map under $g$ to either 0 or 1. Therefore, $g = \mathbb{1}_B$ and so $f(B) = \mathbb{1}_B = g$.

This shows that $\{0, 1\}$ is a subobject classifier for $\mathsf{Set}$.

If in a set $S$ we equip it by the equality relation $\sim \coloneqq =$ then the category $\mathsf{C}$ whose objects are $\mathsf{Obj}(\mathsf{C}) = S$ and morphisms are simply identities $(a, a)$

for every object $a \in S$ (all the morphisms are identities because no $a \sim b$ for distinct $a$ and $b$in $S$). This is a catetgory in which every morphism is an isomorphism because theyre all identity. ■

**Exercise. aluffi** Suppose we are given $n$ different morphisms $f_1, \ldots, f_n$ so that $f_k \circ f_{k+1}$ is always well defined for $k = 1, \ldots, n-1$. Then, any valid expression involving parenthesis (, ), elements $f_{1,} \ldots, f_k$ and the composition operator $\circ$ is always independent of the placement of the parenthesis.

**Solution.** Let us denote by $E_k$ a well formed expression of length $k$. By this we mean that there are $k$ morphisms $f_1, \ldots, f_k$ in $E_k$, $k-1$ composition operators $\circ$ and an arbitrary number of appropriately nested parenthesis.

It suffices to show that any $E_k$ can be equivalently written in the form $(\ldots(f_1 \circ f_2) \circ f_3) \circ f_4) \circ \ldots) \circ f_{k-1}) \circ f_k$ which allows us to speak of the unique legal expression involving $k$ morphisms $f_1, \ldots, f_k$ and we can simply write that as $f_1 \circ f_2 \circ \cdots \circ f_k$.

We proceed by strong induction on $n$.

For the base case of $n = 3$, $f_1 \circ (f_2 \circ f_3) = (f_1 \circ f_2) \circ f_3$ is simply the associativity of $\circ$ in $\mathsf{C}$.

Our induction hypothesis is that $K_k = (\ldots(f_1 \circ f_2) \circ f_3) \circ f_4) \circ \ldots) \circ f_{n-1}) \circ f_n$ for all expressions $K_k$ of length $k \in \mathbb{Z}_{>0}$ with $3 \leqslant k \leqslant n$.

Now let $K_{n+1}$ have length $n+1$. Then, there are $l, m \in \mathbb{Z}_{>0}$ such that $l + m = n + 1$ and expressions $K_l$ and $K_m$ such that $K_{n+1} = K_l \circ K_m$. Now, there are two cases:

Case 1: $m = 1$. Then, $l = n$, $K_m = f_{n+1}$. So, by the induction hypothesis

$$K_l = (\ldots(f_1 \circ f_2) \circ \ldots) \circ f_{n-1}) \circ f_n$$

Consequently,

$$K_{n+1} = (\ldots(f_1 \circ f_2) \circ \ldots) \circ f_{n-1}) \circ f_n) \circ f_{n+1}.$$

Case 2: $m > 1$. By the induction hypothesis, $K_m$ can be written in the form $K_m = K_{m-1} \circ f_{n+1}$, and so

$$K_{n+1} = K_l \circ (K_{m-1} \circ f_{n+1}) = (K_l \circ K_{m-1}) \circ f_{n+1}$$

But, $K_l \circ K_{m-1}$ is itself an expression of length $n$, so, by the induction hypothesis,

$$K_l \circ K_{m-1} = (\dots(f_1 \circ f_2) \circ f_3) \circ \dots) \circ f_n$$

Finally,

$$K_{n+1} = (\dots(f_1 \circ f_2) \circ f_3) \circ \dots) \circ f_n) \circ f_{n+1}$$

which completes the induction step. ∎

**Exercise. aluffi** Given a set $S$ and a relation $\sim$ defined on $S$, define the category $\mathsf{C}$ as such: $\mathsf{Obj}(\mathsf{C}) = S$ and given objects $a, b \in S$,if $a \sim b$ then there is a morphism $a \xrightarrow{f} b \in \mathsf{Hom}_{\mathsf{C}}(a, b)$ given by $f = (a, b)$ and so $\mathsf{Hom}_{\mathsf{C}}(a, b) = \{(a, b)\}$ and if $a$ and $b$ are not related by $\sim$ then $\mathsf{Hom}_{\mathsf{C}}(a, b) = \varnothing$. For what conditions on $\sim$ is C a groupoid?

**Solution.** We claim that if $\sim$ is an equivalence relation then $\mathsf{C}$ is a groupoid. Indeed, we saw that $\mathsf{C}$ is a category if $\sim$ is reflexive and transitive. If $\sim$ is also symmetric, that is, if for any $a, b \in S$, $a \sim b$ then $b \sim a$. Indeed in this case, every morphism $f : a \to b$ has the left inverse $g : b \to a$ given as $(b, a)$ so that $gf = (b, a)(a, b) = (a, a) = 1_a$. Similarly, $f : a \to b$ has the right inverse $g = (b, a)$ so that $(a, b)(b, a) = (b, b) = 1_b$. ∎

**Exercise. aluffi** Let $A, B$ be objects of a category $\mathsf{C}$, and let $f \in \mathsf{Hom}_{\mathsf{C}}(A, B)$ be a morphism.

- Prove that if $f$ has a right inverse, then $f$ is an epimorphism.
- Find an epimorphism without a right inverse.

**Solution.** Suppose a morphism $f : A \to B$ between objects $A, B \in \mathsf{Obj}(\mathsf{C})$ has a right inverse $g : B \to A$ so that $fg = 1_B$. Then, Fix an object $Z$ and a pair of morphisms $\alpha', \alpha'' : B \to \mathsf{Z}$ in $\mathsf{C}$ so that $\alpha' \circ f = \alpha'' \circ f$. This implies that $\alpha' \circ f \circ g = \alpha'' \circ f \circ g$ which implies that $\alpha' \circ 1_B = \alpha'' \circ 1_B$ and therefore $\alpha' = \alpha''$.

Just take the morphism $(2, 3)$ in the category given in Eg-3.3. Any right inverse should make it so that $(2, 3) \circ g = (3, 3)$ so only $(3, 2)$ is a choice but this is not a morphism that makes sense in this category. ■

**Exercise. aluffi** Can you define a subcategory $\mathsf{C}_{\text{nonmono}}$ of $\mathsf{C}$ by restricting to morphisms that are *not* monomorphisms?

**Solution.** If $\mathsf{C}$ is the category in Eg-3.3 then every morphism is *monic.* In that case the set $\mathsf{Hom}_{\mathsf{C}_{\text{nonmono}}}(a, a)$ for any $a \in \mathsf{Obj}(\mathsf{C})$ is always empty because even $1_A$ is monic.

It might be worth thinking about whether we can do this if we do admit $1_A$ to every $\mathsf{Hom}_{\mathsf{C}_{\text{nonmono}}}(a, a)$ in $\mathsf{C}_{\text{nonmono}}$. Perhaps we might have to look at some examples like $\mathsf{Grp}$ or $\mathsf{CRing}$.

**Exercise. aluffi** Give a concrete description of mono and epi morphisms in the category $\mathsf{MSet}$ you constructed in Ex-3.9.

**Solution.** TODO

# References

[Paolo Aluffi(2009)]